

Factors Predictive of Outcome in Childhood Epilepsy

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ABSTRACT

To identify early predictive factors of outcome in childhood epilepsy, the case records of all children with new-onset epilepsy presenting to a single neurology practice over a 10-year interval were reviewed. Only children with more than 2 years of follow-up were included. Cox regression analysis was used to identify factors predictive of remission (successful cessation of medication). One hundred ninety-six children (mean age 7.6 ± 3.7 years at first seizure, mean follow-up 55 ± 30 months) were identified. Ninety-eight of 196 children (50%) had an idiopathic epilepsy, 63 of 196 (32.1%) had cryptogenic epilepsy, and 35 of 196 (17.9%) had remote symptomatic epilepsy. At final assessment, 52.6% were in remission, 12.8% had a poor outcome (recurrent seizures on therapeutic antiepileptic drug levels within 6 months prior to the final assessment), and 6.9% were intractable (more than one seizure/month over 1 year with failure of three or more anticonvulsants). One year after initiating treatment, factors associated with a lower probability of remission included seizure recurrence in the 6- to 12-month interval after therapy initiation (hazard ratio 0.24), multiple seizure types (hazard ratio 0.40), and mental retardation at onset (hazard ratio 0.19). Factors predictive of a poor outcome included seizure recurrence in the 6- to 12-month interval after therapy initiation (odds ratio 21.6), more than one seizure type (odds ratio 8.9), and global developmental delay at onset (odds ratio 8.9). Factors predictive of intractability included multiple seizure types (hazard ratio 6.5), mental retardation at onset (hazard ratio 7.2), and seizure recurrence in the first 6 to 12 months of treatment (hazard ratio 70). It appears that response in the first 6 to 12 months on antiepileptic medication is predictive of outcome. Clinical features of the underlying epilepsy and concurrent neurologic conditions were independently associated with intractability and a lower probability of remission. (*J Child Neurol* 2005;20:898–904).

Epilepsy is a heterogeneous group of neurologic conditions characterized by recurrent, unprovoked seizures.¹ The prevalence of childhood epilepsy has been estimated at 3.5 to 7.2 per 1000 children,² making it one of the most common neurologic disorders in pediatrics. Fortunately, the majority of children with epilepsy will

become seizure free as adults, with remission rates increasing with length of follow-up.³

Studies on epilepsy prognosis have contributed greatly to our understanding of the natural history of the disease and shaped our current practice. Before the 1980s, four seizure-free years were traditionally required before discontinuing medication. However, as a result of prospective studies, current practice usually considers two seizure-free years as sufficient.^{4–6} Studies on epilepsy prognosis, for the most part, fall into two broad categories: predicting remission and predicting intractability. The first outcome is easiest to define: the child responds to medical therapy and seizures abate. Various definitions of remission have nevertheless been used in studies, all requiring a certain seizure-free period (from 6 months to 5 years), with some studies excluding those still on medication at last follow-up.^{7–9} Most studies do not take into consideration possible surgical treatment of epilepsy in their remission group.

More controversial is the definition of intractability. Various features used to define intractability have been seizure frequency and the duration and failure of medical therapy. Factors predict-

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ing intractability have varied considerably in different studies. It is likely that differences in defining intractability, the actual study design, and the selection of the study population have contributed to these differing findings. Certain age groups (children with seizure onset within the first year), seizure types (infantile spasms, myoclonic, atonic), and underlying etiologies (neurodegenerative disorders) are associated with a worse outcome.^{10–12} Long-term follow-up studies of childhood epilepsy have consistently shown that remission rates, even in initially intractable cases, increase with the length of follow-up.^{3,13} Despite these differences, factors predictive of intractability have included remote symptomatic etiology, mental retardation, failure to respond to the initial antiepileptic drug, and initial seizure frequency.^{9–11}

Common questions asked at diagnosis by the parents of an epileptic child include “Will my child continue to have seizures? If my child responds well to medication initially, will he or she continue to do so? Will my child be able to eventually come off medication?” Previous studies have identified conflicting risk factors, and their application to our current population has not been evaluated.

In clinical practice, it would be desirable to better predict the clinical course of a child's epilepsy within a short period after diagnosis. The current study aimed to identify early predictive factors of outcome in all children with new-onset epilepsy requiring anticonvulsant medication in an ambulatory pediatric neurology practice. Our secondary objective was to see whether these factors varied depending on the etiology of a child's epilepsy.

MATERIALS AND METHODS

Subjects

The subjects were identified from a comprehensive computerized database of a single pediatric neurologist (M.I.S.). This relational database modified from Microsoft Access (Microsoft, Redmond, WA) contains demographic, clinical, diagnostic, and treatment information (14 fields total) on all patients seen. Information is entered into this database by the neurologist at the time of initial consultation and updated subsequently at follow-up visits and as the results of investigations become available. Patients entered into this database were seen by the neurologist in four different clinics: a hospital-based private neurology office (1991–1996), a hospital-based ambulatory clinic (together with resident house staff, 1991–ongoing), a hospital-based neonatal neurology clinic for at-risk neonatal intensive care unit survivors (1991–ongoing), and a suburban private pediatric neurology clinic (1990–ongoing). Children with new-onset epilepsy aged between 2 and 17 years were identified through a systematic search of the above database. The medical charts of all such potential subjects were retrieved and systematically reviewed by one investigator (M.O.) in the spring of 2003.

Inclusion and Exclusion Criteria

All subjects who met the following inclusion criteria were included in the study: (1) initial intake and neurologic assessment between July 1, 1991 and June 30, 2000; (2) epilepsy onset between 2 and 17 years of age; (3) treatment with an antiepileptic drug; and (4) a minimum follow-up of 24 months. Patients who only had febrile seizures or a single unprovoked seizure or were not seen at the onset of their epilepsy, either seen as a second opinion or for transfer of care, were excluded from the study.

Intake and Definitions

The inpatient and outpatient charts of the identified subjects were systematically reviewed using a structured format for identification of clinical

features at intake, etiologic determination, clinical course, and known outcome. The clinical features at intake recorded were gender, age at first afebrile seizure, seizure type, the number of seizures prior to treatment, family history of epilepsy, a history of febrile seizures, perinatal history, comorbid conditions, neurologic examination, and the most abnormal electroencephalogram (EEG) before initiation of treatment.

The seizure type was classified according to the classification of the International League Against Epilepsy.¹⁴ The perinatal history was considered normal if the child was delivered at term without complications requiring neonatal intensive care unit observation. The comorbid conditions present at onset included the presence of developmental delay (global, pure motor, or isolated speech delay), mental retardation, attention-deficit hyperactivity disorder (ADHD), and learning disabilities. Global developmental delay was defined as a significant delay in two or more of the following developmental domains: gross or fine motor, speech and language, social and interpersonal, and activities of daily living.¹⁵ ADHD and learning disability were defined according to their respective *Diagnostic and Statistical Manual of Mental Disorders-IV* criteria.¹⁶ Mental retardation was diagnosed on the basis of clinical judgment supplemented by neuropsychologic evaluation when available. The following features on neurologic examination were recorded: microcephaly or macrocephaly, dysmorphic features, pyramidal tract signs (diplegia, hemiplegia, quadriplegia), and developmental delay. The most abnormal EEG recorded before treatment was reviewed for the following features: the presence of abnormal slowing (focal or generalized), epileptiform activity (focal, multifocal, or generalized) and the presence of epileptiform activity triggered by hyperventilation, photic stimulation, or sleep. An EEG was considered abnormal if there was the presence of any slowing or epileptiform activity. In our institution, an EEG is frequently performed even after a first seizure. If the initial EEG is normal, it is routinely repeated.

Etiologic determination was divided into three broad categories: idiopathic epilepsy, remote symptomatic epilepsy, or cryptogenic epilepsy. A child was considered to have epilepsy if he or she had two or more epileptic seizures in the absence of any immediate identified cause.¹⁷ If an antiepileptic drug was started after a single seizure, a child was considered to have epilepsy only if seizures recurred. Idiopathic epilepsies were identified according to the International League Against Epilepsy definition.¹⁸ Symptomatic epilepsy was defined as seizures occurring with an identifiable previous brain injury or in the presence of a static encephalopathy known to be associated with an increased risk of seizures. Symptomatic categories included cerebrovascular causes, acquired cerebral trauma, cerebral dysgenesis, central nervous system tumors, and previous central nervous system infections. Cryptogenic epilepsy was defined as unprovoked seizures occurring without a disorder or abnormality associated with an increased risk of epilepsy.

Features of the clinical course recorded included the following: initial antiepileptic drug; subsequent antiepileptic drugs; the need to add an antiepileptic drug because of poor seizure control; seizure recurrence in the first 3, 6, and 12 months of treatment; the use of nonmedicinal regimens, such as the ketogenic diet or neurosurgery; and the need to restart an antiepileptic drug after discontinuation. The context in which seizures recurred was also recorded: recurrence only off medication or on therapeutic antiepileptic drug levels. The outcome recorded included number of months of follow-up since initiation of an antiepileptic drug, treatment with an antiepileptic drug at the time of last follow-up, and the number of months seizure free.

Remission was defined as successful cessation of medication. It was standard practice to discontinue medication after a seizure-free interval of 24 months. The last intake date for the study was June 2000, and the patients'

charts were reviewed 33 to 35 months later, allowing for a greater possible follow-up interval. Any child who completed 2 years of follow-up was considered to be eligible for the study. A child who had seizures at intake, was commenced on medication, and had no further seizures over the next 2 years was reviewed at the end of the 2-year period and placed on a regimen for tapering medication. The child was considered to meet the criteria for remission at this point. A subsequent review often did not occur unless there were recurrent seizures or other medical issues. All children were in the study for a period long enough to meet the criteria for remission. Should seizures recur, the child no longer met the criteria for remission. Thus, children who went into remission had a shorter mean duration of follow-up than those who did not achieve remission. Subjects who were tapered off medication after a neurosurgical intervention were excluded from this group because such an invasive procedure is not performed unless medical therapy has failed to control the child's often intractable seizures and would not fit the profile for a "good outcome" for parents at the onset of their child's epilepsy. A poor outcome was defined as having recurrent seizures on therapeutic and antiepileptic drug levels within 6 months prior to the final assessment or having required epilepsy surgery or the use of the ketogenic diet. Intractability was defined as seizures recurring at least monthly for more than a year associated with failure of at least three antiepileptic drugs at the time of final assessment. For those children in remission at the time of their final assessment, the interval between initiation of antiepileptic drug treatment and successful cessation of medication was also recorded (time to remission).

Statistical Analyses

Statistical analysis was performed using *Statistical Package for Social Sciences* software, version 11.5 (SPSS Inc, Chicago, IL, 2002). Differences in proportions between the children studied and those lost to follow-up were compared using the chi-square test. Differences in age distributions between these groups were compared using an independent-sample *t*-test. Factors predictive of remission were modeled using Cox regression analysis. The time to remission was taken as the interval between initiation of medication and remission (successful cessation of medication). Patients who were not in remission at the time of their final assessment were censored at this time. The presence or absence of a given finding at study intake was modeled as a binary variable. Cox regression analysis was used to compensate for the bias toward remission associated with a longer duration of follow-up (given the tendency of epilepsy to spontaneously remit). For Cox regression analyses, the hazard ratio is quoted followed by the 95% confidence interval. Kaplan-Meier plots were constructed to illustrate the likelihood of remission associated with specific variables at intake. For situations in which the assumptions associated with Cox regression analysis did not hold, survival curves were compared using the log-rank test.

Logistic regression analyses were used to assess factors predictive of a poor outcome or intractability at outcome. In this case, the duration of follow-up was incorporated into the model as a covariate to compensate for the tendency for spontaneous improvement associated with longer durations of follow-up. The variables entered into the Cox regression and logistic regression models were gender; age at onset of seizures; abnormal perinatal history; family history of epilepsy; developmental delay at onset; mental retardation at onset; learning disability at onset; ADHD at onset; pyramidal tract signs at initial examination; number of seizures prior to treatment; EEG abnormality; cryptogenic epilepsy; remote symptomatic epilepsy; the presence of an epilepsy syndrome as defined by the International League Against Epilepsy; seizure recurrence within 3, 6, and 12 months and within the 6- to 12-month interval after initiating therapy; and the number of seizure types. Two separate analyses were performed to identify factors 3 and 12 months after initiating medication that predicted outcome.

RESULTS

The records of 437 children in the study were reviewed. One hundred ninety-seven children did not meet the inclusion criteria: 57 had an age at onset under 2 years or over 17 years; 66 had an intake date after June 30, 2000; 24 were seen as a second opinion or for transfer of care; and 16 did not have a final diagnosis of epilepsy. Thirty-four children started on an antiepileptic drug after a single seizure never had a recurrence and were excluded. Forty-four children were excluded because their duration of follow-up was less than 2 years. Of these, one patient with dysgerminoma died of complications from gliomatosis cerebri, five patients moved, and five patients were from northern Quebec and were not seen regularly in follow-up. The clinical features of the remaining 33 children lost to follow-up and the 196 who met all inclusion criteria are summarized in Table 1. Age at first seizure, gender distribution, and seizure type at epilepsy onset did not differ significantly between the group of children included in the study and those lost to follow-up. However, there was a significant difference ($P < .001$) in the number of children within the three etiologic groups.

The clinical features of our study group at the time of the initial diagnosis are outlined in Tables 2 and 3. The characteristics of their clinical course are outlined in Table 4. Medication was discontinued in 121 subjects (61.7%), of whom 21 (10.7%) were restarted on medication owing to eventual seizure recurrence. Of these, three were eventually successfully withdrawn from medication a second time. The mean duration of follow-up was 55 ± 30 months (range 24–163 months). The outcome at the time of final assessment is summarized in Table 5. The majority of children (103 of 196; 52.6%) were in remission, with only a minority meeting the criteria for a poor outcome (25 of 196; 12.8%) or intractability (12 of 196; 6.9%). In the separate causal groups, remote symptomatic epilepsy carried the worse prognosis, with 28 of 35 children (80%) not in remission at the last assessment. The other two groups were comparable, with 37 of 98 children (37.8%) with idiopathic epilepsy and 28 of 63 children (44.4%) with cryptogenic epilepsy not being in remission at the time of the last assessment. In all three groups, remission rates increased with length of follow-up. These results are illustrated in a Kaplan-Meier plot (Figure 1).

Predictors of a Lower Probability of Remission

Factors that independently predicted a lower probability of remission were identified using Cox regression analysis. Three months after initiating an antiepileptic drug therapy, the factors associated with a lower probability of remission were mental retardation at onset (hazard ratio 0.23; 95% confidence interval 0.07–0.74), remote symptomatic epilepsy (hazard ratio 0.46; 95% confidence interval 0.23–0.92), and having more than one seizure type (hazard ratio 0.34; 95% confidence interval 0.15–0.78). Factors associated with a lower probability of remission 1 year after initiating antiepileptic drug therapy were mental retardation at onset (hazard ratio 0.19; 95% confidence interval 0.06–0.59), recurrent seizures in the 6- to 12-month interval after therapy initiation (hazard ratio 0.24; 95% confidence interval 0.10–0.60), and having more than one seizure type (hazard ratio 0.40; 95% confidence interval 0.18–0.93). The seizure frequency before treatment was not predictive of outcome, even when children with absence seizures were eliminated from the analysis.

Table 1. Clinical Features of Subjects Included and Those Lost to Follow-Up

	<i>Subjects in the Study</i>		<i>Lost to Follow-Up Subjects</i>	
	n	%	n	%
Age at first seizure, mean $\pm$ SD (yr)	7.6 $\pm$ 3.7		7.7 $\pm$ 4.1	
Gender				
Male	98	50.0	17	51.5
Female	98	50.0	16	48.5
First seizure type				
Generalized absence	38	19.4	3	9.1
Generalized tonic-clonic	38	19.4	9	27.3
Partial secondarily generalized	53	27.0	8	24.2
Partial simple or complex	58	29.6	13	39.4
Other	9	4.6	0	0.0
Initial medication prescribed				
Carbamazepine	80	40.8	15	45.5
Valproic acid	85	43.4	9	27.3
Phenytoin	15	7.7	8	24.2
Clobazam	8	4.1	1	3.0
Other	8	4.1	0	0.0
Etiology				
Cryptogenic	63	32.1	27	81.8*
Remote symptomatic	35	17.9	1	3.0*
Idiopathic	98	50.0	5	15.2*
Childhood absence epilepsy	26	13.3	3	9.1
Benign occipital epilepsy	4	2.0	1	3.0
Benign rolandic epilepsy	41	20.9	1	3.0
Others	27	13.8	0	0.0

*Reaching statistical significance.

In the separate causal groups, the following factors were significantly associated with a lower probability of remission: children with idiopathic epilepsy: more than one seizure type (hazard ratio 0.19; 95% confidence interval 0.05–0.81), an abnormal EEG (hazard ratio 0.26; 95% confidence interval 0.09–0.76), and the presence of a learning disability (hazard ratio 0.44; 95% confidence interval 0.19–0.99) and cryptogenic epilepsy: seizure recurrence in the 6- to 12-month interval after therapy initiation (hazard ratio 0.13;

95% confidence interval 0.03–0.54) and developmental delay at onset (hazard ratio 0.31; 95% confidence interval 0.12–0.78). In children with remote symptomatic epilepsy, both seizure recurrence in the 6- to 12-month interval after the initiation of medication (log rank 5.65; $P = .02$) and seizure recurrence in the first 12 months (log rank 14.2; $P < .001$) were significantly associated with a lower probability of achieving remission. In fact, no child with seizure recurrence in the 6- to 12-month interval after therapy initiation achieved remission.

Table 2. Clinical Features at Diagnosis

<i>Variable</i>	<i>Subjects</i>	
	n	%
Gender		
Male	98	50.0
Female	98	50.0
Number of seizures prior to treatment		
1	42	21.4
2	45	23
3–5	41	20.9
6–10	9	4.6
> 10	59	30.1
Neurologic findings		
Abnormal examination	49	25.0
Mental retardation	24	12.2
Developmental delay	42	21.4
ADHD	28	14.3
Learning disability	33	16.8
Features on history		
Abnormal perinatal history	24	12.2
Prior febrile seizure	27	13.8
Typical febrile seizure	18	9.2
Atypical febrile seizure	9	4.6
Positive family history of epilepsy	71	36.2
First-degree relative	19	9.7
Second-degree relative	52	26.5

ADHD = attention-deficit hyperactivity disorder.

Predictors of a Poor Outcome

Multivariate factors that independently predicted a poor outcome were determined using logistic regression. Factors predictive of a poor outcome 3 months after initiating antiepileptic drug therapy

Table 3. Epilepsy Groups

<i>Variable</i>	<i>Subjects</i>	
	n	%
Epilepsy groups		
Idiopathic epilepsy	98	50.0
Cryptogenic epilepsy	63	32.1
Remote symptomatic epilepsy	35	17.9
Idiopathic epilepsy		
Benign rolandic	41	20.9
Childhood absence	26	13.3
Juvenile myoclonic	11	5.6
Juvenile absence	8	4.1
Benign occipital	4	2.0
Idiopathic generalized tonic-clonic	6	3.1
Primary photosensitive	2	1.0
Remote symptomatic epilepsy		
Cerebral dysgenesis	15	7.6
Cerebrovascular event	15	7.6
Head trauma	4	2.0
Other (Angelman syndrome)	1	0.5

Table 4. Characteristics of the Clinical Course

Variable	Subjects	
	n	%
Number of seizure types		
1	169	86.2
2	15	7.7
≥ 3	12	6.1
Status epilepticus	21	10.7
Early recurrence of seizures		
Within 3 mo of treatment	52	26.5
Within 6 mo of treatment	65	33.1
Between 6 and 12 mo of treatment	29	14.8
Need to add second AED	53	27
Change AED	45	23.0
Number of AED tried		
1	114	58.2
2	49	25.0
3	23	11.7
≥ 4	10	5.1

AED = antiepileptic drug.

were more than one seizure type (odds ratio 14.7; 95% confidence interval 4.7–46.1) and global developmental delay at onset (odds ratio 3.3; 95% confidence interval 1.1–10.2). Children classified as having idiopathic epilepsy were less likely to have a poor outcome (odds ratio 0.13; 95% confidence interval 0.03–0.52). One year after initiating treatment, the following factors were predictive of a poor outcome: seizure recurrence in the 6- to 12-month interval after therapy initiation (odds ratio 21.6; 95% confidence interval 6.3–74.0), more than one seizure type (odds ratio 8.9; 95% confidence interval 2.6–31.2), and global developmental delay at onset (odds ratio 8.9; 95% confidence interval 2.4–32.7).

Predictors of Intractability

Three months after initiating antiepileptic drug therapy, logistic regression analysis identified the presence of more than one seizure type (odds ratio 17.4; 95% confidence interval 4.8–63.1) as being predictive of intractable epilepsy at outcome. One year after initiating treatment, the presence of more than one seizure type (odds ratio 6.5; 95% confidence interval 1.9–35.4), the presence of mental retardation (odds ratio 7.2; 95% confidence interval 1.0–50.8), and seizure recurrence in the 6- to 12-month interval after initiating therapy (odds ratio 70.4; 95% confidence interval 7.5–661.4) were predictive of intractable epilepsy at outcome.

DISCUSSION

Childhood epilepsy in our clinical practice had a generally favorable outcome, with over half of children achieving remission without needing neurosurgical intervention. Previous studies have reported that 40% to 50% of children will eventually become seizure free and be able to cease medication.^{19–21} Inherent in our calculation of remission rates are two assumptions. The first is that children with seizure recurrence returned to the evaluating neurologist (M.I.S.) or to our institution (either another neurologist or to the emergency room). In Montreal, the average waiting list for a pediatric neurologist is 1 year. The hospital records were reviewed for any subsequent visits to the hospital. It is therefore probable that the vast majority of children with recurrent seizures were recognized as not having attained remission. The second assumption is

Table 5. Outcome at Last Follow-Up

Variable	Subjects	
	N	%
Medication status		
Off medication	107	54.6
On 1 medication	63	32.1
On > 2 medications	26	13.3
Seizure control (time since last seizure, mo)		
< 1	12	6.1
1–6	17	8.7
6–12	15	7.7
12–24	40	20.4
> 24	112	57.1
Nonmedicinal treatment		
Epilepsy surgery	6	3.1
Ketogenic diet	2	0.1
Poor outcome (6 mo)	25	12.8
Intractable epilepsy	12	6.9
Remission	103	52.6

that a child who was in remission at the end of the study subsequently remained in remission. The remission rate for children who had less than 1 year between achieving remission and data evaluation could be an overestimation. However, only four children evaluated in 2000 achieved remission, and charts were reviewed 33 to 36 months later. Conversely, the remission rate could be underestimated for children whose intake was in 2000, assuming that they might have attained remission if given more than 33 months of follow-up. Eleven children fall into this group, with six being seizure free for 24 months or more at the last follow-up but not tapered off medication, one being seizure free for 12 to 24 months, and four being seizure free for less than 12 months.

Our study population was drawn from a single practice in one of two pediatric neurology referral centers in Montreal, Quebec. Although our group does not encompass all children seen in our city with new-onset epilepsy, it is a representative sample. Within our hospital, all neurologists see children with new-onset epilepsy

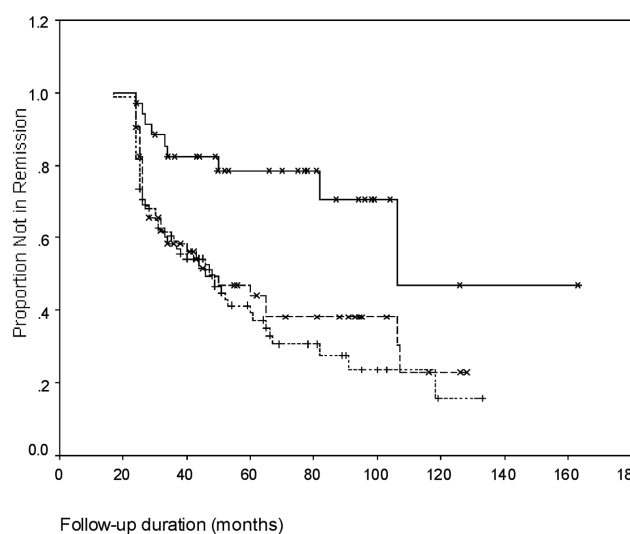


Figure 1. Kaplan-Meier curve illustrating relapse rates in cohort over time according to type of epilepsy.

and provide ongoing follow-up. Locally, general pediatricians do not provide management for seizure disorders. The referral pattern between the two centers and among the clinicians within our center has not been evaluated previously, but we see no reason for differences to exist. During the 10-year study period, there was no significant change in clinical practice. All children were evaluated by the same physician (M.I.S.) in a similar fashion. The choice of medication and timing of discontinuation (2 years seizure free) were consistent throughout the study period. Each antiepileptic drug was titrated to the highest tolerated therapeutic dosage, beyond which seizure recurrence was considered a "failure" and a second agent was added.

The number of seizures prior to treatment was not predictive of outcome in our study. Many previous studies reported a high pre-treatment seizure frequency as predictive of refractory epilepsy.⁸⁻¹¹ There are two possible explanations for this difference. The first is a difference in our study population and design. Our study included children with all seizure types. Children with absence seizures, often excluded from other studies, usually have multiple daily seizures before being recognized and treated. However, when absence seizures were eliminated from the analysis, the number of seizures prior to treatment remained unresponsive of outcome. A second more plausible explanation for this difference is that anticonvulsant medications are not antiepileptogenic. Although treatment has been shown to lower the risk of recurrence, it does not necessarily change the long-term prognosis of the underlying epilepsy.^{22,23} The number of seizures prior to treatment might simply reflect the natural course of the epilepsy and, depending on the study design, have a positive or negative association with outcome.

Perhaps more significant than the initial seizure frequency prior to treatment is the duration of active epilepsy before achieving control on medication. If seizures are not medically controlled in the first year of treatment, the chance of remission is significantly reduced.^{19,24,25} Sillanpaa and colleagues reported a 30-year follow-up of children with epilepsy in Turku, Finland.⁷ The single best predictor of outcome in their study was a rapid response to therapy, defined as "a reduction in the frequency of seizures of 75 to 100% within three months of beginning treatment." Kwong and colleagues reported a cohort of children with epilepsy in Hong Kong, with a mean follow-up of 5.9 years.²⁶ On multivariate analysis, seizure recurrence in the second 6 months after treatment was found to be predictive of medical intractability. In our study, seizure recurrence in the 6- to 12-month interval after initiating treatment was associated with a lower probability of remission and a higher probability of a poor outcome and intractability. Whether this association reflects the severity of the underlying epilepsy or the epileptogenic effect of persistent seizures remains to be determined.

In our study, multiple seizure types predicted a poor outcome, intractability, and a lower probability of remission. The majority of children had a single seizure type, with only 27 of 196 (13.8%) having more than one seizure type. This group included seizure types widely accepted to be associated with a worse prognosis, such as myoclonic, atonic, and atypical absence seizures.¹⁹⁻²¹ All four children with Lennox-Gastaut syndrome and 11 children with juvenile myoclonic epilepsy were included in this group. Previous studies that have directly explored this association are in agreement with our finding.^{27,28}

The presence of mental retardation at onset was associated with intractability and a lower probability of remission. Global developmental delay at onset was associated with a poor outcome. Previous studies have largely agreed with this finding.^{8,19,20} It is likely that the presence of a developmental disability does not in itself directly alter outcome but rather reflects the enhanced severity of a more widespread underlying central nervous system abnormality.

Remote symptomatic epilepsy is known to carry a worse prognosis.²⁹ In our study, remote symptomatic epilepsy was associated with a lower probability of remission and children with idiopathic epilepsy were less likely to have a poor outcome. Sillanpaa reported that the probability of remission in children with a favorable early response to treatment was 92% and 67% for idiopathic and cryptogenic epilepsies, respectively, but only 30% for those with a remote symptomatic epilepsy.²¹ Consistent with these findings, both the remote symptomatic and cryptogenic epilepsies showed seizure recurrence in the first 6 to 12 months of treatment to be predictive of outcome. In fact, in the symptomatic epilepsy group, none of the children with seizure recurrence in the first 6 to 12 months of treatment reached remission. Among children with idiopathic epilepsy, factors predictive of nonremission included the presence of a learning disability at onset, an abnormal EEG, and the occurrence of multiple seizure types. The latter selects for children with juvenile myoclonic epilepsy, who have very low remission rates.^{30,31} Idiopathic epilepsies with a single seizure type, such as benign rolandic epilepsy and benign occipital epilepsy, have higher remission rates.^{32,33} One advantage to establishing the diagnosis of a specific epilepsy syndrome is the ability to better predict a child's clinical course and outcome. Grouping this heterogeneous group of epilepsies yielded three factors predictive of outcome. More syndrome-specific analysis would perhaps be desirable, but the lower number of children in each group would not yield statistically meaningful results from our sample.

The outlook for childhood epilepsies is generally good, and even initially intractable children can become seizure free with time. In clinical practice, it would be desirable to predict within a short period after diagnosis the likely clinical course of the child's epilepsy. Other than providing parents with answers, this could allow for more aggressive medical treatment (rational polytherapy) and earlier consideration of neurosurgical intervention or non-medicinal treatment (ketogenic diet). In our study, the response in the first 6 to 12 months after initiating antiepileptic medication was predictive of outcome. The clinical features of epilepsy (failure to completely control seizures within the first year of treatment, more than one seizure type) and concurrent neurologic conditions (mental retardation, developmental delay, remote symptomatic epilepsy) were independently associated with a lower probability of epilepsy remission. It is important to note, however, that even initially intractable epilepsies can attain remission on longer follow-up.

Although our understanding of the etiology and prognosis of childhood epilepsy has improved over the last two decades, much work lies ahead in this field. The outcome for these children should not simply be measured in terms of seizure recurrence and remission but should also take into account the effects on quality of life, participation, and cognition. Future longitudinal prospective multicenter studies are needed, with attention not only to epilepsy etiology and prognosis but also to diverse measures, which contributes significantly to the overall morbidity in these children.

References

1. Cowan LD: The epidemiology of the epilepsies in children. *Ment Retard Dev Disabil Res Rev* 2002;8:171–181.
2. Kozyskyj AL, Prasad AN: The burden of seizures in Manitoba children: A population-based study. *Can J Neurol Sci* 2004;31:48–52.
3. Huttenlocher PR, Hapke RJ: A follow-up study of intractable seizures in childhood. *Ann Neurol* 1990;28:699–705.
4. Shinnar S, Vining EPG, Millits ED, et al: Discontinuing antiepileptic medication in children with epilepsy after two years without seizure. *N Engl J Med* 1985;313:976–980.
5. Arts WFM, Visser LH, Loonen MCB, et al: Follow-up of 146 children with epilepsy after withdrawal of antiepileptic therapy. *Epilepsia* 1988;29:244–250.
6. Bouma PAD, Peters ACB, Arts RJHM, et al: Discontinuation of antiepileptic therapy: A prospective study in children. *J Neurol Neurosurg Psychiatry* 1987;50:1579–1583.
7. Sillanpaa M, Jalava M, Kaleva O, Shinnar S: Long-term prognosis of seizures with onset in childhood. *N Engl J Med* 1998;338:1715–1722.
8. Camfield C, Camfield P, Gordon K, et al: Outcome of childhood epilepsy: A population-based study with a simple predictive scoring system for those treated with medication. *J Pediatr* 1993;122:861–868.
9. Kwan P, Brodie MJ: Early identification of refractory epilepsy. *N Engl J Med* 2000;342:314–319.
10. Sillanpaa M: Remission of seizures and predictors of intractability in long-term follow-up. *Epilepsia* 1993;34:930–936.
11. Berg AT, Shinnar S, Levy SR, et al: Early development of intractable epilepsy in children: A prospective study. *Neurology* 2001;56:1445–1452.
12. Chawla S, Aneja S, Kashyap R, Mallika V: Etiology and clinical predictors of intractable epilepsy. *Pediatr Neurol* 2002;27:186–191.
13. Camfield P, Camfield C: Childhood epilepsy: What is the evidence for what we think and what we do? *J Child Neurol* 2003;18:272–287.
14. Commission on Classification and Terminology of the International League Against Epilepsy: Proposal for revised clinical and electroencephalographic classification of epileptic seizures. *Epilepsia* 1981;22:489–501.
15. Majnemer A, Shevell MI: The diagnostic yield of the neurologic assessment of the developmentally delayed child. *J Pediatr* 1995;127:193–199.
16. American Psychiatric Association: *Diagnostic and Statistical Manual of Mental Disorders*, 4th ed. Washington, DC, American Psychiatric Association, 1994.
17. Commission on Epidemiology and Prognosis, International League Against Epilepsy: Guidelines for epidemiologic studies on epilepsy. *Epilepsia* 1993;34:592–596.
18. Commission on Classification and Terminology of the International League Against Epilepsy: Proposal for a revised classification of epilepsies and epileptic syndromes. *Epilepsia* 1989;30:389–399.
19. Annegers JF, Hauser WA, Elveback LR: Remission of seizures and relapse in patients with epilepsy. *Epilepsia* 1979;20:729–737.
20. Eriksson KJ, Koivikko MJ: Prevalence, classification, and severity of epilepsy and epileptic syndromes in children. *Epilepsia* 1997;38:1275–1282.
21. Sillanpaa M: Long-term outcome of epilepsy. *Epileptic Disord* 2000;2:79–88.
22. Musicco M, Beghi E, Solari A, et al: Treatment of first tonic-clonic seizure does not improve prognosis of epilepsy. *Neurology* 1997;49:991–998.
23. Shinnar S, Berg AT: Does antiepileptic drug therapy prevent the development of “chronic” epilepsy? *Epilepsia* 1996;37:701–708.
24. Reynolds EH: The prevention of chronic epilepsy. *Epilepsia* 1988;29(Suppl 1):S25–S28.
25. Arts WHM, Geerts AT, Brouwer OH, et al: The early prognosis of epilepsy in childhood: The prediction of a poor outcome. The Dutch study of epilepsy in childhood. *Epilepsia* 1999;40:726–734.
26. Kwong KL, Sung WY, Wong SN, So KT: Early predictors of medical intractability in childhood epilepsy. *Pediatr Neurol* 2003;29:46–52.
27. Beghi E, Tognoni G, Collaborative Group for the Study of Epilepsy: Prognosis of epilepsy in newly diagnosed referred patients: A multicenter prospective study. *Epilepsia* 1988;29:236–243.
28. Brorson LO, Wranne L: Long-term prognosis of childhood epilepsy: Survival and seizure prognosis. *Epilepsia* 1987;28:324–330.
29. Shinnar S, Pellock JM: Update on the epidemiology and prognosis of pediatric epilepsy. *J Child Neurol* 2002;17(Suppl 4):S4–S17.
30. Oguni H, Hayashi K, Osawa M: Long-term prognosis of Lennox-Gastaut syndrome. *Epilepsia* 1996;37(Suppl 3):S44–S47.
31. Panayiotopoulos CP, Obeid T, Tahan AR: Juvenile myoclonic epilepsy: A 5-year prospective study. *Epilepsia* 1994;35:285–296.
32. Bouma PAD, Bovenkerk AC, Westendorp RGJ, Brouwer OF: The course of benign partial epilepsy of childhood with centro-temporal spikes: A meta-analysis. *Neurology* 1997;48:430–437.
33. Ferrie CD, Beaumanoir A, Guerrini R, et al: Early-onset benign occipital seizure susceptibility syndrome. *Epilepsia* 1997;38:285–293.